Project Plan

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Project Plan

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1. Project Name
    a. Setting up a company LAN network with an Apache web server
2. Project Description
    a. A LAN network will be built for a private company that utilizes an internal web server, set up on an Ubuntu machine, for sending company related information and a web page with the directory of employees will be displayed. The entire network will be secured using Access Control Lists (ACLs), encrypted passwords, and a firewall. A Kali Linux machine will be set up to penetration test (pen test) the network and exploit vulnerabilities of the Ubuntu web server. In order to do so, the Kali machine will have to use a port scanner and vulnerability scanner. The pen testing tools that the Kali Linux machine will be using are John the Ripper, Nmap, and Metasploit. Pen testing is used to find weaknesses in the network and machines to fix them before they cause damage. Testing will be conducted using the ping command to ensure that all devices are connected.
3. Location of Work
    a. At my apartment
4. Equipment
    a. Three Cisco IOU L3 157-3 2018 routers
    b. One Cisco IOU L2 15.2 2019 switch
    c. Kali Linux machine

d. 3 Basic PCs in GNS3 with PuTTY

e. Ubuntu Server running Apache2 service

5. Detailed Objectives

    a. Research

        i. Different ways Metasploit can penetrate the network

        ii. Different options/flags for use with the Nmap command

        iii. John the Ripper tools and utilities

            1. How it works

        iv. Known vulnerabilities for ubuntu web server

            1. Common Vulnerabilities and Exploits (CVEs)

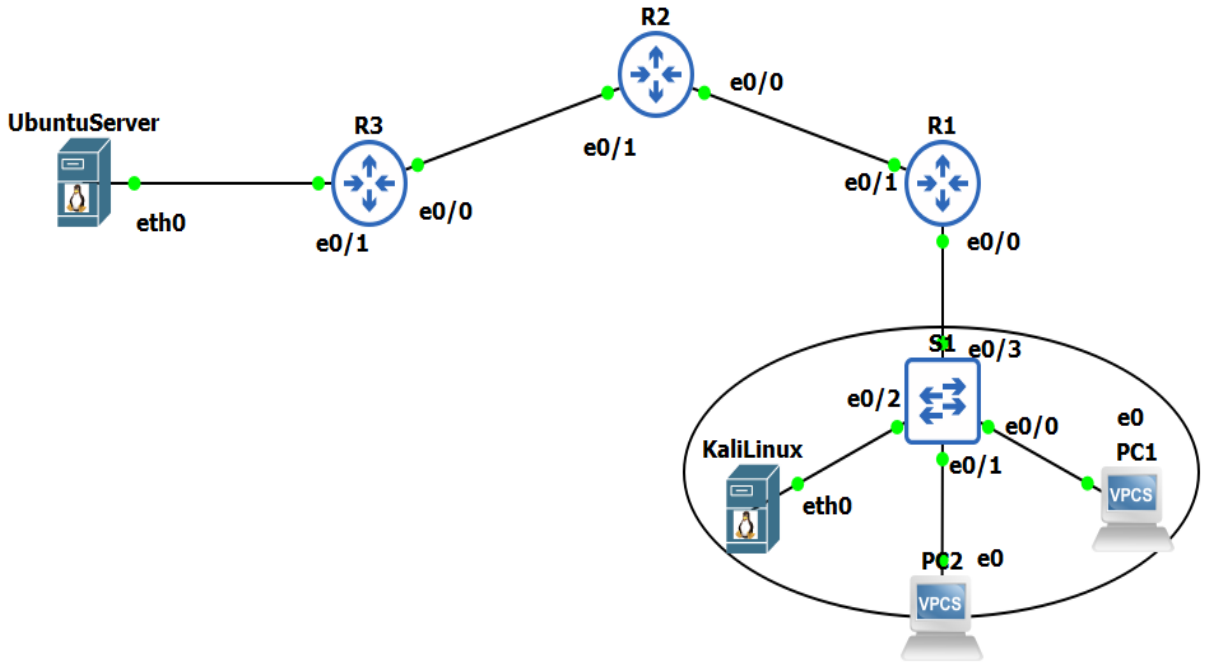
        v. Encryption methods of user's passwords

        vi. Web server basic security

    b. Design

        i. Topology and addressing table

            1. Each device in the topology will be virtualized using GNS3 software. GNS3 include images for the Cisco Routers and switches.



Device	Interface	IP Address	Netmask	Gateway
R1	Eth0/0	192.168.10.1	255.255.255.0	N/A
	Eth0/0.10	192.168.10.65	255.255.255.248	N/A
	Eth0/1	192.168.20.1	255.255.255.0	N/A
R2	Eth0/0	192.168.20.2	255.255.255.0	N/A
	Eth0/1	192.168.30.1	255.255.255.0	N/A
DHCP	Eth0/2	192.168.83.137	255.255.255.0	192.168.83.2
R3	Eth0/0	192.168.30.2	255.255.255.0	N/A
	Eth0/1	192.168.40.1	255.255.255.0	N/A
S1	Vlan10	192.168.10.66	255.255.255.248	192.168.10.65
Ubuntu Server	Eth0	192.168.40.10	255.255.255.0	192.168.40.1
Kali Machine	Eth0	192.168.10.69	255.255.255.248	192.168.10.65
PC1	Eth0	192.168.10.67	255.255.255.248	192.168.10.65
PC2	Eth0	192.168.10.68	255.255.255.248	192.168.10.65

ii. Operating systems used

1. Debian Linux

    a. Kali Linux

    b. Ubuntu

iii. Tools used to attack the network

1. John the Ripper – a command line interface (CLI) tool used to crack passwords
2. Nmap – a CLI reconnaissance tool used to map the entire network and search for open ports
3. Metasploit – a pen testing tool used at the command line to exploit a machine

c. Implementation

i. Routers

1. Using a routing protocol to establish connection, OSPF
2. Configuring Access Control Lists for inbound and outbound traffic
    a. There will be 3 ACLs created
3. Disabling unused ports to prevent an attack from a port scan
4. A firewall will be configured to at router 2 to prevent inbound traffic that is not trusted
5. Prevent login attacks by encrypting plain text passwords and requiring log in at the console line

6. Creating strong passwords by requiring a minimum character amount
7. Enable Secure Shell (SSH) to enable secure remote logins
8. Inter-VLAN routing

ii. Switch

1. Disabling unused ports to prevent an attack from a port scan
2. Configuring switchport security by only allowing two MAC addresses to be learned automatically
3. Prevent log in attacks by encrypting plain text passwords and enabling log in at the console line
4. Creating strong passwords by requiring a minimum character amount
5. Trunking of VLAN10

iii. Ubuntu Web Server

1. Download and install the apache2 service
2. Configure the apache2 service by making an HTML file the index to display the web page

iv. Workstation

1. Configure default settings such as the IP address and default gateway
2. On the same network as the Kali Linux machine

v. Kali Linux

1. Penetration testing tools
    a. John the Ripper, Metasploit, and Nmap
2. Pen test the network using the scanning tools to discover vulnerabilities

d. Testing
    i. Connectivity
        1. Ping command to test connectivity from PC1 to R2 and from PC1 to R3
        2. Verify the network is running correctly
    ii. Penetration testing
        1. John the Ripper
        2. Metasploit
        3. Nmap
    iii. Ubuntu Server
        1. Ensure connection to online web page
        2. Ensure that login credentials are secure and encrypted

e. Documentation
    i. Project Proposal
    ii. Project Analysis
    iii. Project Description
        1. Router configuration
        2. Switch configuration

3. Ubuntu server configuration
4. Kali server configuration
5. Penetration testing
6. Metasploit options
7. John the Ripper
8. Nmap options

iv. Project Presentation

v. Weekly Project Reports

vi. References